

LLM-as-a-Judge Report: Self-doubt GSM8K

What was measured?

The judge scores visible self-questioning in the reasoning trace, independently of whether the final answer is correct. A score of 0 means a straight-line trace with no hesitation; 10 means pervasive uncertainty, reversals, or looping. The judge returns the score, a verbatim evidence quote when available, and a one-line summary. The tables below use 0–10 scores and include only valid answered traces from `gsm8k_first2_5.jsonl`.

Examples at different scores

The examples below show the score, final-answer outcome, the judge’s evidence, and its one-line summary. An empty evidence field means that the trace contained no suitable verbatim cue for that dimension.

Trace `typo25_real10`, **index** 54. **Score:** 0 on the 0–10 scale. **Final-answer outcome:** wrong. **Judge evidence:**

This equation is always true, indicating that the number of kittens adopted (x) can be any non-negative integer.

One-line summary: The trace led to the scores because it provided a clear and direct solution to the problem without any noticeable hesitation or confusion.

Trace `clean`, **index** 8. **Score:** 2 on the 0–10 scale. **Final-answer outcome:** correct. **Judge evidence:**

Wait a second, let me double-check to make sure I didn’t make any mistakes

One-line summary: The trace led to the scores due to a brief moment of hesitation and double-checking, as seen in the phrase ‘Wait a second, let me double-check to make sure I didn’t make any mistakes’.

Trace `typo50_real40`, **index** 19. **Score:** 5 on the 0–10 scale. **Final-answer outcome:** correct. **Judge evidence:**

Wait a second, 6 mph seems pretty fast for hiking. Is that correct?

One-line summary: The model’s repeated checks and second-guessing, as seen in phrases like ‘Wait a second, 6 mph seems pretty fast for hiking. Is that correct?’, led to the scores.

Trace typo25_real40, index 7. Score: 8 on the 0–10 scale. **Final-answer outcome:** wrong. **Judge evidence:**

Wait, hold on. The problem says 'Ten Carla as to recant the download from the beginning.' Hmm, maybe that was a typo or something.

One-line summary: The trace is marked by frequent re-evaluations and second-guessing, particularly when encountering typos such as 'Ten Carla as to recant the download from the beginning' and 'how load does it take'.

Aggregate tables

Table 1: Mean score by real-word typo ratio

real_ratio	n	self_doubt_score_mean
10	1447	1.001
40	1457	0.889
70	1455	0.826

Table 2: Mean score by typo percentage

typo_percentage	n	self_doubt_score_mean
0	494	0.684
25	1450	1.052
50	1454	0.865
75	1455	0.799

Table 3: Mean score by final-answer correctness

outcome	n	self_doubt_score_mean
correct	3267	0.852
wrong	1586	0.946

Consequences

In this run, self-doubt decreases as the real-word ratio increases: the mean is 1.001 at 10 percent real-word typos, 0.889 at 40 percent, and 0.826 at 70 percent. This is consistent with the interpretation that non-real-word typos are more visibly strange and can trigger more doubt, while real-word typos can be read fluently and silently.